

SECTION 1: OPERATORS (1–25)

1. What is the output?

```
let a = 10;  
let b = ++a + a++;  
console.log(b);
```

2. Predict the output:

```
let x = 5;  
console.log(x++ + ++x);
```

3. What will be printed?

```
let a = 10;  
console.log(a++ + ++a + a);
```

4. Find the output:

```
let x = 10;  
let y = x++ * 2;  
console.log(x, y);
```

5. Output?

```
let a = 5;  
let b = a-- + --a;  
console.log(b);
```

6. What is the result?

```
console.log(10 + "5" - 2);
```

7. Output?

```
console.log("10" - "5" + 2);
```

8. Output?

```
console.log("5" * "2" + 1);
```

9. Output?

```
console.log(10 > 5 > 2);
```

10. Output?

```
console.log(10 == "10");
```

11. Output?

```
console.log(10 === "10");
```

12. What will this print?
`console.log(true + false);`

13. Output?
`console.log(false == 0);`

14. Output?
`console.log(null == undefined);`

15. Output?
`console.log(5 & 1);`

16. Output?
`console.log(5 | 1);`

17. Output?
`console.log(5 ^ 1);`

18. Output?
`console.log(2 << 3);`

19. Output?
`console.log(16 >> 2);`

20. Output?
`let a = 10;`
`a += a -= 5;`
`console.log(a);`

21. Output?
`console.log(!!"hello");`

22. Output?
`console.log(!"");`

23. Output?
`console.log(5 && 0 && 10);`

24. Output?
`console.log(0 || "" || "JS");`

25. Output?
`console.log(typeof (5 + "5"));`

SECTION 2: CONDITIONS (26–50)

26. Check if a number is positive, negative, or zero.
27. Check if a number is even or odd (no % operator).
28. Find the largest of 2 numbers.
29. Find the largest of 3 numbers using if–else.
30. Check if a year is a leap year.
31. Check if a character is a vowel or consonant.
32. Check if a character is uppercase or lowercase.
33. Check if a number is divisible by 3 and 5.
34. Check if a number is prime (no array).
35. Check if a number is palindrome.
36. Check if a string is empty or not.
37. Check if a string length is greater than 10.
38. Convert marks into grade using if-else.
39. Check if a number lies between 10 and 50.
40. Check if a character is a digit or alphabet.
41. Find absolute value without Math.abs().
42. Check if a number is a perfect square.
43. Check if the string starts with a vowel.
44. Check if the string ends with a consonant.
45. Check login:
`username === "admin" && password === "1234"`
46. Use switch to print day name from day number.

- 47. Use switch to print month days.
- 48. Check if a number is multiple of 7.
- 49. Compare two strings lexicographically.
- 50. Check if a number is a strong number.

SECTION 3: LOOPS (51–75)

- 51. Print numbers from 1 to 10.
- 52. Print numbers from 10 to 1.
- 53. Print even numbers 1–50.
- 54. Print odd numbers 1–50.
- 55. Print table of 5.
- 56. Print factorial of a number.
- 57. Find the sum of digits of a number.
- 58. Reverse a number using a loop.
- 59. Count digits in a number.
- 60. Check if a number is palindrome using a loop.
- 61. Print Fibonacci series up to n terms.
- 62. Find the power of a number (a^b) using a loop.
- 63. Print all divisors of a number.
- 64. Count even and odd digits in a number.
- 65. Print prime numbers between 1–100.
- 66. Find GCD of two numbers.
- 67. Find LCM of two numbers.

68. Print pattern:

```
*  
**  
***  
****
```

69. Print pattern:

```
1  
12  
123  
1234
```

70. Print sum of first n natural numbers.

71. Print reverse counting skipping multiples of 3.

72. Find the sum of squares 1 to n.

73. Print numbers from 1–100, but replace:

- multiples of 3 → "Fizz"
- multiples of 5 → "Buzz"
- multiples of both → "FizzBuzz"

74. Check Armstrong number.

75. Print digits of numbers line by line.

SECTION 4: STRINGS & STRING METHODS (76–100)

76. Find length of a string without .length.

77. Reverse a string using a loop.

78. Check if the string is palindrome.

79. Count vowels in a string.

80. Count consonants in a string.

81. Count spaces in a string.
82. Convert lowercase to uppercase (manual logic).
83. Convert uppercase to lowercase.
84. Find the first occurrence of a character.
85. Find the last occurrence of a character.
86. Count occurrences of a character.
87. Remove spaces from a string.
88. Replace all vowels with *.
89. Check if the string contains only digits.
90. Check if the string contains only alphabets.
91. Capitalize the first letter of the string.
92. Convert sentence to title case.
93. Find the longest word in a string.
94. Remove duplicate characters.
95. Find frequency of each character (no object).
96. Compare two strings without ==.
97. Check if the string contains substring.
98. Split string manually by space.
99. Reverse words in a sentence.
100. Check if the string is an anagram.